Operating Systems
Week 7
Lecture/Discussion 7

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Previous

- Scheduling Algorithms (SJF Preemptive- Round Robin)

Today Discussion

- Before Break Scheduling Algorithms Quiz
- After break Lecture Continue

Quiz

1. FCFS

Process	Burst Time
P_1	24
P_2	3
P_3	3

Suppose that the processes arrive in the order: P_1 , P_2 , P_3
Draw a Gantt chart by using FCFS

2. Priority Scheduling

Process	Burst Time	Priority
P_1	10	3
P_2	1	1
P_3	2	4
P_4	1	5
P_5	5	2
P_6	6	6

Draw Priority scheduling Gantt Chart
Waiting time of each Process?
Average waiting time?

Priority-based Scheduling

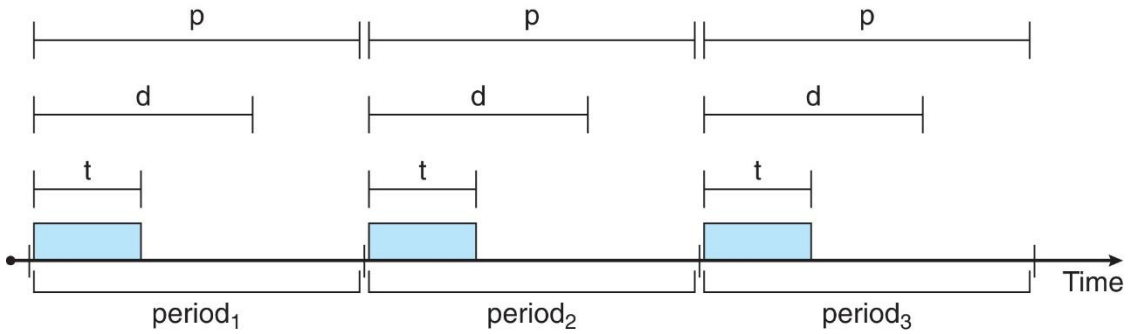
For real-time scheduling, scheduler must support preemptive, priority-based scheduling

- But only guarantees soft real-time

For hard real-time must also provide ability to meet deadlines

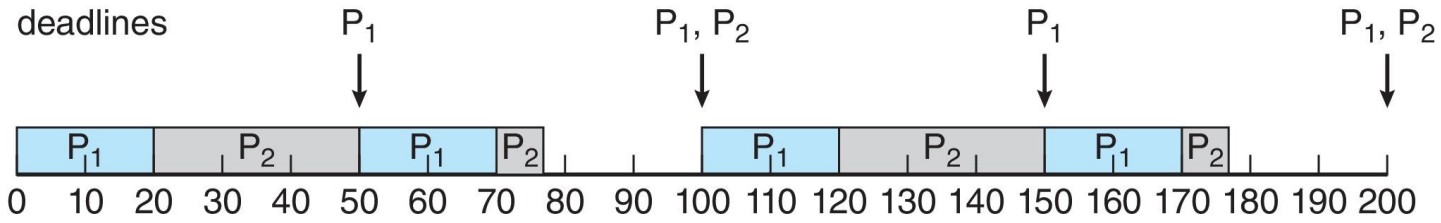
Processes have new characteristics: **periodic** ones require CPU at constant intervals

- Has processing time *t*, deadline *d,* period *p*
- $0 \le t \le d \le p$
- **Rate** of periodic task is $1/p$



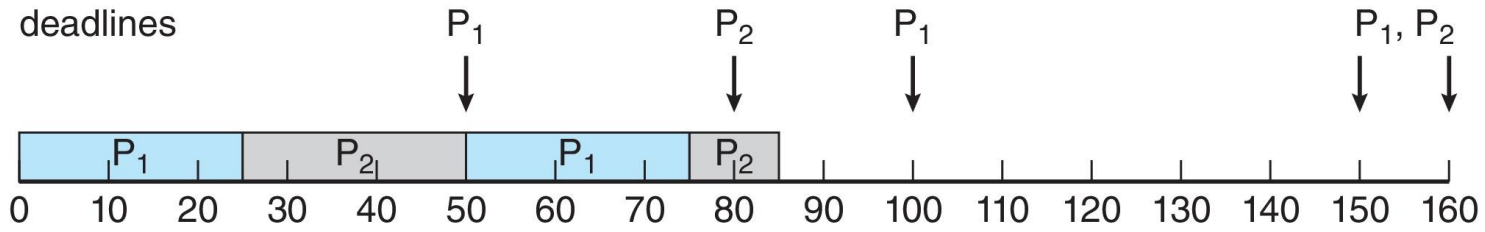
Rate Monotonic Scheduling

A priority is assigned based on the inverse of its period
Shorter periods = higher priority;
Longer periods = lower priority
P_1 is assigned a higher priority than P_2.



Missed Deadlines with Rate Monotonic Scheduling

Process P_2 misses finishing its deadline at time 80
Figure

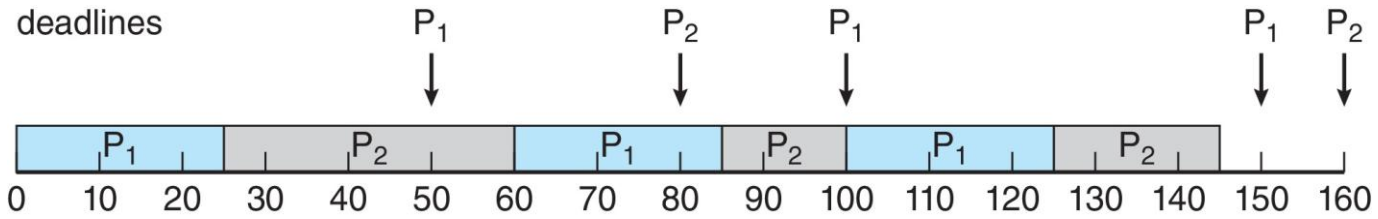


Earliest Deadline First Scheduling (EDF)

Priorities are assigned according to deadlines:

- The earlier the deadline, the higher the priority
- The later the deadline, the lower the priority

Figure



Proportional Share Scheduling

T shares are allocated among all processes in the system
An application receives *N* shares where *N* < *T*
This ensures each application will receive ***N*** / *T* of the total processor time

Windows Scheduling

Windows uses priority-based preemptive scheduling
Highest-priority thread runs next
Dispatcher is scheduler
Thread runs until (1) blocks, (2) uses time slice, (3) preempted by higher-priority thread
Real-time threads can preempt non-real-time
32-level priority scheme
Variable class is 1-15, **real-time class** is 16-31
Priority 0 is memory-management thread
Queue for each priority
If no run-able thread, runs **idle thread**

Windows Priority Classes

Win32 API identifies several priority classes to which a process can belong

- REALTIME_PRIORITY_CLASS, HIGH_PRIORITY_CLASS, ABOVE_NORMAL_PRIORITY_CLASS,NORMAL_PRIORITY_CLASS, BELOW_NORMAL_PRIORITY_CLASS, IDLE_PRIORITY_CLASS
- All are variable except REALTIME

A thread within a given priority class has a relative priority

- TIME_CRITICAL, HIGHEST, ABOVE_NORMAL, NORMAL, BELOW_NORMAL, LOWEST, IDLE

Priority class and relative priority combine to give numeric priority

Base priority is NORMAL within the class

If quantum expires, priority lowered, but never below base

Windows Priority Classes (Cont.)

If wait occurs, priority boosted depending on what was waited for

Foreground window given 3x priority boost

Windows 7 added **user-mode scheduling** (**UMS**)

- Applications create and manage threads independent of kernel
- For large number of threads, much more efficient
- UMS schedulers come from programming language libraries like
- C++ **Concurrent Runtime** (ConcRT) framework

Windows Priorities

	real-time	high	above normal	normal	below normal	idle priority
time-critical	31	15	15	15	15	15
highest	26	15	12	10	8	6
above normal	25	14	11	9	7	5
normal	24	13	10	8	6	4
below normal	23	12	9	7	5	3
lowest	22	11	8	6	4	2
idle	16	1	1	1	1	1